

RatherKnow — Product & Technical Specification

Version 1.1.0 · as built, June 2026 · INTERNAL

This document supersedes `rk-1.1.0-PRD.md` and `rk-1.1.0-TRD.md`, which describe intent. This one describes what exists. Every figure in Parts I and II was read out of the running code, not from a plan. Part III is the defect register. Part IV is the forward spec for 1.2.0.

Not for publication. Part III names shipped defects, placeholder reliabilities and promises made on the site that have nothing behind them yet. The publishable artefacts are `frontend/public/docs/ratherknow-scoring-spec.pdf` and `frontend/public/docs/ratherknow-archetypes.pdf`.

Part 0 · Summary

RatherKnow is a **single-player** instrument. It measures how one person chooses in relationships, from that person's own answers, and it never describes anybody else.

Five instruments and one reflection are live behind an account. **456 items** in total. Two instruments are established with calibrated norms; three are ours and developmental. Nothing is purchasable: the pricing ladder is published, and the checkout does not exist.

The thing that distinguishes the product commercially is not the psychometrics — it is the refusals. No compatibility score, no types, no predictions, no matching, a published evidence tier on every instrument, immutable version-stamped results, and the equations printed in the report. That set is difficult to copy because it costs conversion, and it is the reason a practitioner can put their name beside it.

At a glance	Value
Live instruments	5 + Flag Check
Total items	456
Backend tests	185, green
Revenue	£0 / \$0 — no payment integration exists
Deployment	preview only; no production deployment evidenced

Part I · Product, as built

1. Premise and refusals

You answer instruments about yourself. The product tells you how you choose: the speed of it, which qualities you over-weight, which you cannot see because they come free to you, and where the gap sits between who you are and who you say you want.

The refusals are load-bearing product decisions, not marketing. In order of how expensive they are to keep:

1. No compatibility score. Not between people, not between archetypes, not ever.
2. No matching, no profiles, no other people's data.
3. No types, boxes or quadrants — positions between poles, with both poles named.
4. No predictions about how any relationship goes.
5. No urgency, no countdowns, nothing that expires.
6. Every instrument publishes its own evidence tier, including the weak ones.
7. Composite equations are published, including their uncalibrated gains.
8. Results are version-stamped and never recomputed.

9. Safety and abuse referrals on every page, free, behind nothing.
10. Stated at the top of the promise: it cannot find you love. Nothing can.

Copy expressing 1–10 is hash-locked (§18).

2. Users, accounts and situations

One user type. Registration requires name, email, password and a **relationship situation**:

Value	Label
single_dating	Single and dating
in_relationship	In a relationship
post_breakup	Post-breakup

The situation reframes interpretation copy and the PDF notes. **It never changes a score.** It is user-editable at any time, and changing it produces a new report snapshot rather than rewriting the old one (§14).

Access is gated **before questions begin** — deliberate, because results are retrievable later and an anonymous result cannot be. The earlier "Free. Anonymous. No account, no card." claim was rewritten rather than left standing.

The Flag Check is the sole exception: no account, by design.

3. The instruments

Key	Name	Items	Tier	Scale	Stated time
essential	Essential Mirror	100 (50 × 2 lenses)	Developmental	1–5 Likert	~25 min
personality	Personality Mirror	130	Established	1–5 Likert	15–20 min
eq	EI Mirror	140	Established	1–5 Likert	15–20 min
closeness	Closeness Mirror (MI-AS-36)	37	Developmental	1–7, from bank	~5 min
everyday	Everyday Mirror (MI-EV-49)	49	Developmental	forced choice, 2 options	~7 min

Essential — six archetypes scored twice, once as yourself and once as the partner you want. Produces the Delta. Full internals and defects in docs/ARCHETYPES_SPEC.md.

Personality — 15 primary factors and 5 global dimensions. The item bank is 130 of the P150's items (Switch module removed), scored against the same norms snapshot as the parent product, so the two are directly comparable.

EI — Goleman four-domain, sub-dimension roll-ups, min/max normalised.

Closeness — two dimensions (anxiety, avoidance) on a well-replicated research tradition, with our own items. Both are **distance** scores: high means more of the thing named, not more ease.

Everyday — the only instrument that asks you to *choose* rather than describe yourself, and therefore the only one that can contradict your self-description. 28 position items (four per domain across pace, sociability, order, novelty, money, recovery, togetherness) then 21 round-robin priority comparisons. Carries the only behavioural validity evidence in the product: circular-triad consistency ($\zeta = 1 - 24d/336$), side bias, and a per-item time floor.

Personality and EI item order is deterministically **interleaved** so no trait or sub-dimension repeats consecutively. Closeness keeps its spec order. Everyday keeps a fixed order for everyone with **sides randomised per session** — required, because the side-bias index is meaningless otherwise.

4. What a reader gets

Four layers, in increasing distance from the raw data:

1. **Scores** — frozen, immutable, version-stamped.
2. **Position language** — commonness fractions on normed factors; positions between named poles elsewhere. No sten and no "x out of 10" reaches a reader.
3. **"How you choose"** — per-instrument translation of scores into selection behaviour. The product's actual value proposition.
4. **Cross-check** — where instruments agree (signal), where they disagree (a finding), and a synthesis. A bare "no tensions worth reporting" is not an acceptable terminal state.

Plus a printable single-instrument PDF and a combined PDF across everything finished.

5. The Delta

The distance between the self lens and the ideal lens on the Essential Mirror. The one number that says something neither lens says alone. A zero Delta on the archetype named as the ideal partner is a finding, not a contradiction, and the report says so explicitly.

6. Flag Check

Nine questions — eight items plus a safety item — about one specific relationship. Free, no account, ~3 minutes. Stored in its own collection, `mirror_v2_reflections`, never alongside scored sessions.

It is never scored, banded, labelled, risk-rated or diagnosed. It categorises responses (`reprice / explain / late / none`) and reflects them back. The safety item routes to referrals. This constraint is absolute and structurally enforced by keeping it out of the scoring path entirely.

7. Content surface

Public	Gated
<code>/ /promise /methodology /safety /learn /learn/:slug /faq /samples /flag-check /archetypes /archetypes/:slug /partners /register /login /forgot-password /reset-password</code>	<code>/take/:instrument /results/:sessionId /mirrors</code>

Header nav is five items: The instruments, Archetypes, Promise, Learn, Safety. Archetype pages are menu-reachable — a standing requirement.

8. Commercial model

Published ladder, **USD**:

Tier	Price	Adds	Live?
Your Archetype	Free	Essential Mirror, archetype, Delta	Yes
The Everyday Reading	\$14	Everyday Mirror	No
The Full Reading	\$29	Personality, EI, Closeness, read together	No

\$29 is stated on the page as the **ceiling for the whole practice**. Nothing renews and nothing expires. Both paid tiers are labelled *not yet purchasable*, because no payment integration exists and the page would otherwise be lying.

The guarantee — "if your results don't produce enough for us to say something useful, we'll tell you, and refund half" — is published in body copy. See §15 and Part III/R1: the mechanism that would decide when it fires currently fires on essentially every reading.

9. Partner programme

`/partners`, linked from the footer sitewide. Eight sections, leading with the ten refusals rather than the terms. Offer: 30% for twelve months from first visit, no exclusivity, no lock-in, free access for the practitioner.

Applications: four fields plus a name → `POST /api/partners/apply`, open (no account needed), 3-per-24h rate limit, stored in `partner_applications`.

The page promises a partner link and a dashboard. Neither exists (Part III/R2).

Part II · Technical, as built

10. Architecture

React 19 SPA + FastAPI + MongoDB (Motor), supervisor-managed, single pod behind a Kubernetes ingress. Backend on `0.0.0.0:8001`, frontend on `3000`; everything under `/api` routes to the backend. No queue, no cache, no worker, no object storage.

```
frontend/src/{pages,components,content,lib}
backend/{server.py,auth.py,routes/,services/,constants/}
docs/  scripts/  tests/  memory/
```

11. Data model

Collection	Holds	Immutable?
<code>users</code>	account, bcrypt hash, situation	mutable
<code>mirror_v2_sessions</code>	in-progress responses, <code>presentation.side_map</code> , completed <code>result</code>	result written once
<code>mirror_v2_reflections</code>	Flag Check responses — never scored	append
<code>results</code>	canonical scored snapshot, <code>algo_version</code> , <code>user_id</code>	<code>\$setOnInsert</code>
<code>narratives</code>	resolved layer-3 narrative per (session, <code>display_version</code> , situation)	<code>\$setOnInsert</code>
<code>rendered_pdfs</code>	PDF bytes per (session, kind, <code>display_version</code> , situation)	write-once
<code>password_resets</code>	reset tokens, 60-min TTL	consumed
<code>login_attempts</code>	lockout counters	rolling
<code>partner_applications</code>	practitioner applications, <code>status: unread</code>	append
<code>status_checks</code>	CRA boilerplate, unused	—

No `PyObjectId/BaseDocument` abstraction: documents use application-generated `uuid4` string ids and `_id` is projected out at every read, so raw `ObjectIds` never reach a response.

12. API surface

`/api/v2` — Mirror

Method	Path	Auth
GET	/instruments	open
POST	/assessments	required
GET	/assessments/{id}	owner
PUT	/assessments/{id}/responses	owner
POST	/assessments/{id}/complete	owner
GET	/assessments/{id}/result	owner
GET	/assessments/{id}/report.pdf	owner
GET	/reports/combined.pdf	required
POST	/mirrors/summary, /mirrors/findings	required
POST/GET/PUT/POST	/reflections...	open (Flag Check)
GET	/diagnostics/mrd	required

/api/auth — register, login, me, me/sessions, forgot-password, reset-password, me/situation (PATCH), claim.

/api/partners — audience-options, apply (both open).

Ownership is enforced on every session-scoped endpoint. Cross-user access returns 403, anonymous 401, incomplete-session PDF 409, unknown 404.

13. Scoring and versioning

Hard rule: scoring is frozen. No score, norm, weight or equation changes without a deliberate versioned programme with parity coverage.

Stamp	Value	Governs
ALGO_VERSION	rk-1.0.0	scoring
bank_version	1.0.0	Closeness and Everyday item banks
DISPLAY_VERSION	disp-1.0.0	commonness / position mapping
CONTENT_VERSION	copy-1.1.0	narrative copy
MRD profile_version	mrd-1.0.0	reliability thresholds

Full derivations are published in [docs/SCORING_SPEC.md](#). Two facts worth restating here:

The sten→percentile table shipped wrong at stens 4 and 7 (27/73 instead of 23/77). Corrected, and pinned to $\Phi((\text{sten} - 5.5)/2)$ by test.

The five Personality globals are $5.5 + \Sigma(\text{weight} \times (\text{sten} - 5.5))$ with **gain 1.0, uncalibrated**, then clipped to 1–10. On an extreme profile Independence computes to **11.2** and publishes as 10. The clip is now recorded (clamp_events, raw_precise) and a clamped value is barred from carrying any population statement.

14. Three-layer immutability

Layer 1 responses and layer 2 scores were already immutable. Layer 3 — narrative — is derived at read time from versioned mapping tables that have already needed one correction, so it is snapshotted.

`services/narrative.py`: on first render the resolved narrative is persisted write-once; on first PDF build the bytes are stored and served thereafter. A `display_version` bump affects only reports rendered after it.

The snapshot key includes `situation`. This is the important design point. Freezing exists to stop *us* silently rewriting a delivered document; it must not stop a reader deliberately asking for a different lens. Switching situation produces a new snapshot; switching back returns the original bytes unchanged. The combined roll-up is invalidated when a new instrument finishes; individual reports never are.

15. The MRD engine — shadow mode

`services/mrd.py`. Nothing should describe two scores as different when the instrument cannot tell them apart. $SEM = SD \cdot \sqrt{(1-\alpha)}$; $MRD = z \cdot \sqrt{(SEM^2 + SEM^2)}$ at $z = 1.645$. Four gates: pairwise, flat-profile, distinctiveness, and clustering.

Scale set	SD	α	MRD
Personality primaries	2.0	0.75	2.326 sten
Personality globals	2.0	0.85	1.802 sten
EI sub-dimensions	0.55	0.75	0.640
EI domains	0.5	0.85	0.451
Closeness dimensions	1.1	0.85	0.991

Every α is a literature placeholder. `config()` refuses to load unless `RK_ALLOW_PLACEHOLDER_ALPHA=1` explicitly acknowledges that.

It ships in **shadow**: gates are evaluated and persisted to `result.mrd.suppressions`, and nothing is withheld from any reader. `RK_MRD_MODE=enforce` flips it. See Part III/R1 for why it has not been flipped.

16. Display layer

`services/display.py`. Pure functions, no DB, no prose. Normed primary factors get a rounded population fraction ("about 1 in 4"), never a decimal and never a two-digit percentile. Composites and clamped values raise `NotNormReferenced` rather than degrading to a guess — the refusal is enforced by exception, not by editorial care.

17. Auth and security

bcrypt with per-password salt. JWT HS256, `JWT_SECRET` from env, **30-day** access token. Five failed attempts per email+address → 15-minute lockout via `login_attempts`; the key uses the forwarded chain, because keying on the Kubernetes peer address made every pod share one bucket. Password reset tokens expire in 60 minutes and unknown emails return success (no enumeration).

18. Copy governance

Sixteen locked copy blocks in `frontend/src/content/locked_copy.json`, consumed only via `content/register.js`, hash-enforced by `tests/test_locked_copy.py`. Changing locked copy requires deliberately regenerating the hash register.

`tests/test_copy_lint.py` statically parses the narrative modules and fails on prediction verbs about the reader ("you will", "expect to", "you'll consistently", "leading to"...), on any phrasing describing a person who did not answer, and on a canned paragraph rendering twice in one document. Anchored on "you", so "we will never sell your data" survives.

"validated" is restricted to Personality and EI reference claims.

19. PDFs

ReportLab. `report_pdf.py` builds single-instrument and combined reports: scores, position language, situation notes, "How you choose", composite provenance, safety constraints, and version stamps. Bytes are snapshotted (\$14). Doc PDFs are built from Markdown via `xhtml2pdf` in `scripts/`.

20. Migration and redirects

Legacy `mymirrorreport.com/mirror/*` and `/mirror-index/*` mappings exist as generated artefacts: `frontend/public/_redirects`, `docs/redirects.conf`, `docs/edge/worker.js`, `docs/redirect_map.json`, built and checked by `scripts/build_redirects.py` / `verify_redirects.py`.

Verified locally against the generated rules only. Not verified against a live domain. Invariant: the parent `/mirror` surface must not be removed until real 301s are confirmed in production.

21. Testing

185 tests, green. `pytest -n 2 --dist loadscope`.

Area	File	n
Instrument lifecycles	<code>test_mirror_v2.py</code>	18
Choosing / cross-check / item order	<code>test_iteration6.py</code>	18
PDF reports	<code>test_pdf_report.py</code>	17
Auth	<code>test_auth.py</code>	16
Everyday Mirror	<code>test_everyday.py</code>	15
This release	<code>test_iteration9.py</code>	14
Combined PDF / situation	<code>test_iteration5.py</code>	14
MRD gates	<code>test_mrd.py</code>	10
Reset / situation / redirects	<code>test_iteration3.py</code>	10
Display mapping	<code>test_display.py</code>	8
Archetype docs	<code>test_archetype_docs.py</code>	8
Composites	<code>test_composites.py</code> + API	14
Snapshot immutability	<code>test_immutability.py</code>	4
Copy lint	<code>test_copy_lint.py</code>	3
Locked copy hashes	<code>test_locked_copy.py</code>	2

22. Configuration

Backend: `MONGO_URL`, `DB_NAME`, `CORS_ORIGINS`, `JWT_SECRET`, `EMERGENT_EMAIL_KEY`, `EMAIL_FROM_NAME`, `APP_BASE_URL`, `RK_ALLOW_PLACEHOLDER_ALPHA`, `RK_MRD_MODE`. Frontend: `REACT_APP_BACKEND_URL`.

No defaults anywhere: missing config fails fast at import.

Part III · Defect and risk register

Ordered by what would hurt most. **Nothing in this section is fixed.**

R1 — The guarantee has no working mechanism · SEVERE

The site promises a half refund when the report can't say enough. The MRD gates would decide that, and in shadow mode their suppression rate across ~1,000 gate-eligible stored results is **1.0**. Personality primaries read flat in 197/213 cases; Closeness in 490/490. On the real reference profile — 8 stens of range, six factors at the ceiling — MRD 2.33 folds all fifteen primaries into a **single undifferentiated cluster** whose centroid is the personal mean, so nothing in it is reportable.

Enforcing today would empty the Personality Mirror and trigger the refund on nearly every reading. The thresholds are almost certainly too conservative because α is a guess, not because the profiles are flat. **Do not enforce, and do not promote that guarantee line, until α is measured.**

R2 — Partner promises with nothing behind them · HIGH

`/partners` promises a personal link and a dashboard of clicks, completions and earnings. There is no attribution, no dashboard and no payout mechanism. Applications land in `partner_applications` with **no notification to anyone**. Accepted knowingly on the basis that the first ten partnerships are founder-led — but the first approved partner is owed either the plumbing or a straight explanation, and somebody has to actually read that collection.

R3 — Essential Mirror archetypes are not independent · HIGH

60 scoring slots drawn from a 50-item lens: 13 items double-counted. The **Torchbearer shares 8 of its 10 items** (5 Challenger, 3 Voyager), so it is largely a linear combination of two others and "primary Torchbearer, secondary Challenger" is near-mechanical. The Diplomat shares 4 of 10. Items **7, 34, 36** feed no archetype at all. `diplomat.reverse = [24]` is **inert** — item 24 is a Challenger item with no reversal, so the Challenger score is probably missing an intended reverse. Full detail: `docs/ARCHETYPES_SPEC.md` §4–6. All three fixes change scores, so they need a version bump with old reports keeping their numbers.

R4 — Reliability is unmeasured across the board · HIGH

No α , no test–retest, no factor structure for any of our own instruments. The Personality/EI norms are inherited. This is the single root cause behind R1 and behind the percentile question. Cheapest first step available now: real test–retest between the two same-item P150 sheets.

R5 — Everyday Mirror pre-test outstanding · MEDIUM

No desirability pre-test on the 49 option pairs. Until it runs, unmatched pairs may behave as a Likert scale with extra steps. Disclosed honestly in the payload, the tier statement and the PDF, and the instrument never anchors a reading alone — but that disclosure is a mitigation, not a fix. Needs 20–30 raters; any pair splitting worse than 65/35 gets rewritten. A pilot of $n \geq 200$ would also show whether four items per domain is enough (it may not be — expect undifferentiated domains).

R6 — Production redirects unverified · MEDIUM

Rules generated and locally checked; never run against a live domain. Deleting the parent `/mirror` surface before real 301s are confirmed would drop the inbound traffic this product is inheriting.

R7 — `compatibility` field name · MEDIUM

The Essential result carries a field literally named `compatibility`, backed by `_COMPATIBILITY`. It holds a top-two blend narrative within one lens — no number, no second person — so it does not breach the refusal. But it is named like the one thing permanently ruled out, and it appears in API responses and stored snapshots. Rename to `blend`, carrying the old key for existing snapshots.

R8 — Outbound email receipt unverified · MEDIUM

The reset flow works and the provider returns 202 Accepted. Nobody has confirmed an email actually arriving in an inbox. Verify before relying on password reset in production.

R9 — Unbounded PDF storage · LOW

`rendered_pdfs` grows one document per session per situation per display version, 7–200KB each. Fine now; needs a retention policy or object storage before real traffic.

R10 — No data retention or deletion path · LOW now, blocking for launch

No retention policy, no export, no account deletion, no GDPR/DSAR route. The product holds intimate self-report data about relationships, which is close to the most sensitive category there is. This must exist before a public launch in any consumer jurisdiction.

R11 — Shadow map incomplete · LOW

`adventurer` (the Voyager) is nobody's shadow, so no reader is ever told it is their shadow pull. `rock` is the shadow of two. Decide whether that is editorial or an oversight and record it.

R12 — No production deployment · CONTEXT

Preview only. No custom domain, no monitoring, no backups, no error tracking, no CI runner executing the hash and lint tests automatically — they are enforced by convention today, not by a gate.

Part IV · Next version — 1.2.0

Sequenced so that nothing depends on research that hasn't happened.

Phase A · Honesty debt (no research required)

A1 Rename `compatibility` → `blend`, old key carried in parallel. (R7)

A2 Import-time assert that every `reverse` entry appears in its own archetype's `questions`. Cheap, and stops R3's third defect recurring whatever is decided about the rest.

A3 Partner application notification via the existing Emergent-managed Resend integration, and confirm an email actually arrives — closing R8 in the same move. (R2, R8)

A4 Account deletion + data export. (R10)

A5 Retention policy for `rendered_pdfs` and `narratives`. (R9)

A6 CI gate that runs the locked-copy hash test, copy lint and the full suite. (R12)

A7 Remove the guarantee line from published copy, or reword it to something the current mechanism can honour. This is the highest-value item in Phase A. (R1)

Phase B · Measurement (unblocks everything else)

B1 Compute real test–retest from the two same-item P150 sheets. Cheapest evidence available.

B2 Measure α per scale on live data; replace `reliability_1_0_0.json`, set `placeholder: false`, bump to `mr-d-1.1.0`, remove `RK_ALLOW_PLACEHOLDER_ALPHA` from the environment so the guard becomes real. (R4)

B3 Re-derive MRD thresholds and re-read the shadow suppression rate. Decide then: enforce, loosen to a 68% interval for within-person profile reading, or keep the gates as an internal honesty check that never governs output. Reinstate the guarantee only if the rate supports it.

B4 Everyday Mirror desirability pre-test, then the $n \geq 200$ pilot. (R5)

B5 Essential Mirror factor structure — does the item overlap in R3 actually distort primary assignment, and by how much? Evidence before re-allocation.

Phase C · Scoring v2 (gated on Phase B, ships as `rk-1.1.0` scoring)

Everything here changes numbers, so it ships as one version bump with parity tests, and every existing report keeps its original figures.

C1 Re-allocate the Essential 50 so archetypes have independent items; allocate or retire items 7/34/36; resolve item 24's reversal. (R3)

C2 Calibrate the global composite gains, or state permanently that they are uncalibrated and keep gain 1.0.

C3 Only if B2 supports it: percentiles on normed scales. Not a relabelling of stems — a defined reference distribution and an updated claims register, or nothing.

Phase D · Product

D1 Progressive unlock across sittings, so value lands after each instrument rather than after 456 items.

D2 Everyday Mirror Read 5 (Negotiability) and the priors register from the draft §8.

D3 Shareable single-finding card — no score, no profile, no second person. The Everyday Mirror produces the one line nobody else can print ("you feel strongest about money and you'd trade it first"), and it can travel without breaking a single refusal.

D4 Archetype cost pull-quotes on the public archetype pages.

Phase E · Commercial (gated on D1)

E1 Stripe, test key already in the environment. Tiers as published in §8; ceiling \$29.

E2 `partner_ref` capture and Stripe metadata attribution. Manual payouts for the first ten; no third-party affiliate platform until there is revenue to attribute. (R2)

E3 Production deployment, domain, monitoring, backups.

E4 Verify the legacy 301s against the live domains, **then** retire parent `/mirror`. (R6)

Explicitly not in 1.2.0

Matching, compatibility scoring, couple mode, partner-facing anything, predictions, a partner affiliate platform, or any relaxation of the ten refusals.

Sources: `backend/` and `frontend/src/` as at June 2026; `docs/SCORING_SPEC.md`; `docs/ARCHETYPES_SPEC.md`; `GET /api/v2/diagnostics/mrd`; `memory/PRD.md`; test reports `test_reports/iteration_1..9.json`.